Manifold Ranker

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***Abstract*—**

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I. Introduction

Ranking municipalities in terms of telecommunications Quality of Service (QoS) poses a challenge to regulatory agencies. The ranking is useful not only for diagnostic purposes but for technical and economical evaluation of a given municipality or region.

Defining a ranking strategy is not trivial. One of the problems raised by this challenge are the absence of stablished ground-truth on which cities are better or worse, therefore, evaluating models results depends on contextual knowledge rather than objective comparison. Also, selecting the essential variables to make this analysis is not clear, not only QoS indicators are required to define the strategy but also covariates such as infrastructure and Quality of Experience (QoE) indicators.

Currently, regulatories agencies uses linear composites to stablish QoS scores that are used to ranking municipalities. Brazilian National Telecommunications Agency (ANATEL) defined the Service Quality Index (IQS) through Reference Values Document (DVR), that encapsulates QoS metrics in single valued score to define the telecommunications quality for each municipality.

This pointwise approach lacks a gobal overview of the performance on the municipalities. By not considering a dominance from one over another entities, the ranking became faulty, with scores indistinguishable one from another. Similar entites are attributed near identical values, and the ocurrences are cluttered on narrow regions of the support domain.

To improve distinguishability, one strategy that can be adopted is to compare features from two municipalities and define which one dominates over the other. To find these similarities it is possible to construct a graph, similar entites will be close to each other on the data space.

To asses all the presented challenges, we propose Manifold Ranker, a GNN-based Pareto-inspired model that learns the geometry of the space where the data live in, based on a constructed k-Nearest Neighbors (kNN) graph, and by entity pairwise competition regresses a score value s_i for each municipalities i.

The remainder of the paper is organized as follows. Section II presents the data used in this work and the currently calculated scores. Section III explain the concepts using throughout the development of the methodology applied and presents the developed Manifold Ranker model. Section IV presents the achieved results and a comparison with the currently applied strategies, and discusses them. Section V make some final remarks to the presented work.

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II. Dataset and Baseline Creation

For the dataset construction we acquired QoS and QoE indicators, and infrastructure covariates provided by ANATEL, creating a matrix $\mathbf{X}^{\text{raw}} \in \mathbb{R}^{i \times d}$, where i correspond to the index key and d correspond to the number of features. $\mathbf{X}^{\text{raw}}$ is indexed by two keys: *month*, which identifies the observation period, and *municipality*, the official Brazilian Institute of Geography and Statistics (IBGE) numeric code that uniquely identifies each municipality. Each row therefore corresponds to one municipality–month pair.

As QoS features we selected eight features normalized to $[0, 1]$ with higher values denoting superior performance, IND_1 (voice-call connection rate), IND_2 (voice-call drop rate, complemented), IND_3 (data-connection rate), IND_4 (download and upload speed compliance rate), IND_5 (data-connection latency, complemented), IND_6 (jitter, complemented), IND_7 (packet-loss rate, complemented), and IND_8 (service availability rate). For QoE feature, it was selected the ISG is the General Satisfaction Index, which is a composite of QoE indicators for a given tuple municipality–period. Infrastructure is represented on the dataset by IBC, which is the Brazilian Coverage Index, and the $\text{INF4}_{\text{UP},\{3\text{G},4\text{G},5\text{G}\}}$ and $\text{INF4}_{\text{DL},\{3\text{G},4\text{G},5\text{G}\}}$ indicators, which records the median download speeds over each mobile data technology: 3G, 4G, and 5G.

To establish a baseline comparison with the currently applied metodologies in regulatory agencies, we present the consolidated calculated scores the legacy IQS and the recently revised IQS. Both of them linearly combines the same QoS features selected to use in the model.

A. Legacy IQS

The Anatel legacy composite, IQS_{leg}, aggregates the K normalized technical quality-of-service indicators through a fixed regulator-defined convex combination,

$$\text{IQS}_{\text{leg}} = y_{\max} \sum_{k=1}^{K} w_k^{\text{leg}} \, \text{IND}_k, \tag{1}$$

where $y_{\max} = 10$ is the upper bound of the user satisfaction scale, each $\mathrm{IND}_k \in [0, 1]$ is a normalized technical indicator with higher values denoting superior performance, and the weight vector $w^{\mathrm{leg}} \in \Delta^{K-1}$ lies on the probability simplex $\Delta^{K-1} = \{w \in \mathbb{R}^K : w_k \geq 0, \sum_k w_k = 1\}$. Under the legacy specification, the weights are fixed at $w^{\mathrm{leg}} = (0.1, 0.1, 0.1, 0.2, 0.1, 0.1, 0.1, 0.2)$, assigning greater relevance to IND_4 and IND_8, while distributing equal weight among the remaining indicators.

B. Revised IQS

The Anatel revised composite, $\mathrm{IQS}_{\mathrm{rev}}$, introduced in 2025 through the updated DVR [**?**], replaces the fixed weights with a performance-dependent scheme combined with a coverage bonus. The indicators are first ranked from worst-performing to best-performing within each municipality, and the weight assigned to rank position r is

$$\alpha_r = \frac{1}{r\, H_K}, \tag{2}$$

where $H_K = \sum_{j=1}^{K} 1/j$ is the K-th harmonic number. The revised composite is then defined as

$$\mathrm{IQS}_{\mathrm{rev}} = y_{\max} \sum_{r=1}^{K} \alpha_r \cdot \mathrm{IND}_{\pi(r)} + B_{\mathrm{cov}}, \tag{3}$$

where π is the permutation ordering indicators from worst to best performance, and B_{cov} is a coverage bonus corresponding to a fixed increment of five points, added whenever coverage with 4G or higher technology in the municipality's urban area reaches at least 95%. When $B_{\mathrm{cov}} = 0$, the revised rule reduces to a linear function of the indicators; when $B_{\mathrm{cov}} \neq 0$, it becomes affine. The inverse-harmonic weighting assigns greater importance to the worst-performing indicators, penalizing municipalities with more critical technical dimensions, in contrast to the static weight distribution of the legacy composite.

III. Methodology

Fig. 1 presents an overview of the pipeline process, which will be detailed as follows:

A. Preprocessing

By exploring the selected data that consists the dataset, we verify that it is composed of heavy-tailed, heterogeneous data. Aiming to transform the data with that characteristics into a suitable representation for neural optimization we define a preprocessing step.

To reduce the dynamic range of the features that follows power-law or log-normal-like distributions, common among infrastructure data we apply a Log1p transformation, shown in Eq. 4, for numerical stability this transformations prevents the $\log(0)$ by adding one to the paramater.

$$\mathbf{X}^{\log} = \log(1 + \mathbf{X}^{\mathrm{raw}}). \tag{4}$$

To reduce outliers we apply a Robust scaler, which filter only the date within an interquantile range selected to be $\{0.05, 0.095\}$, Eq. 5 shows the processing.

$$\tilde{x}_{i,j} = \frac{\mathbf{x}_{i,j}^{\log} - \mathrm{median}(\mathbf{x}_{i,j}^{\log})}{Q_{0.95}(\mathbf{x}_{i,j}^{\log}) - Q_{0.05}(\mathbf{x}_{i,j}^{\log})}. \tag{5}$$

From the robust-scaled features, we apply a Standar scaling, Eq. 6 centering the variables into zero mean and unit variance, conditioning for first-orders optimizers, and making

$$\mathbf{X}_{i,j}^{\mathrm{model}} = \frac{\tilde{x}_{i,j} - \mu_j}{\sigma_j}, \tag{6}$$

where μ_j and σ_j are the mean and the standard deviation for the j feature across all municipalities.

Additionally from the $\mathbf{X}^{\mathrm{model}}$, we create a proxy matrix $\mathbf{X}^{\mathrm{pareto}}$ that serves as support for the model training serving as a "compass" general directional guide. $\mathbf{X}^{\mathrm{pareto}}$ is constructed as shown in Eq. 7, by Min-Max scaling the $\mathbf{X}^{\log}$

$$\mathbf{X}_{i,j}^{\mathrm{pareto}} = \frac{\mathbf{x}_{i,j}^{\log} - \min_i \mathbf{x}_{i,j}^{\log}}{\max_i \mathbf{x}_{i,j}^{\log} - \min_i \mathbf{x}_{i,j}^{\log}} \in [0, 1], \tag{7}$$

Then, a scalar proxy utility is defined as the mean across features

$$u_i = \frac{1}{D} \sum_j \mathbf{X}_{i,j}^{\mathrm{pareto}}, \tag{8}$$

this proxy is not a true Pareto ranking; it is a heuristic that proivdes a consistent direction for training in the abscence of labeled rankings.

B. Graph Construction and Manifold Structure

Graphs are mathematical structures capable extracting relationship from data. These relationship enables partial property sharing, having local properties. Geometrically we can visualize this as an estimation of a manifold.

We selected the kNN as the method for constructing the graph. This method calculates the distance between each node and select the k nearest neighbors by using a distance metric, Minkowski in this case, as a result we get a index matrix with the nearest neighbors for each node $\mathcal{I} \in \{1, ..., N\}^{N \times k}$, with N being the number of nodes.

Most implementations of kNN algorithm traces the most nearest neighbor as the node itself, distance zero from itself, i.e., $i \to i$, creating a loop in the graph. This loop brings a residual glance to the model, that can be explored to prevent information vanishing.

C. Positional Encoding

D. Model Architecture

E. Training Objectives and Loss Terms

$$\mathcal{L} = \lambda_{\mathrm{rank}} \mathcal{L}_{\mathrm{rank}} + \lambda_{\mathrm{recon}} \mathcal{L}_{\mathrm{recon}} + \lambda_{\mathrm{geo}} \mathcal{L}_{\mathrm{geo}} + \lambda_{\mathrm{compass}} \mathcal{L}_{\mathrm{compass}} \tag{9}$$

IV. Results and Discussion

V. Conclusions

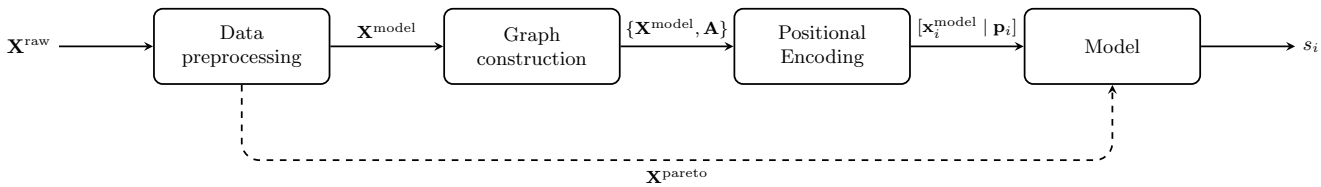


Fig. 1: Pipeline for the proposed methodology

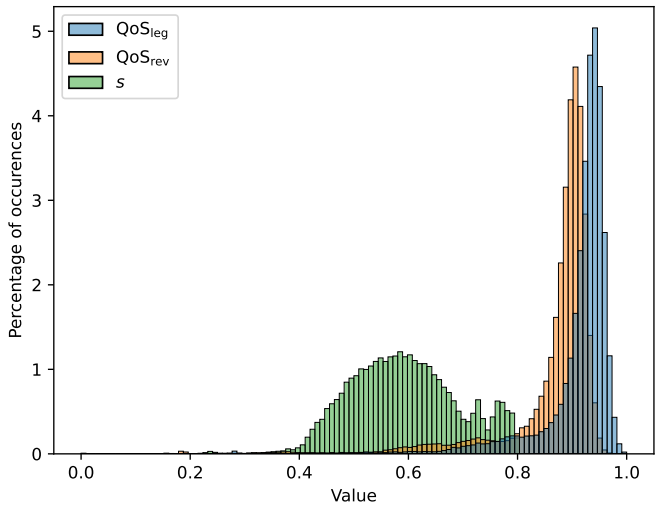
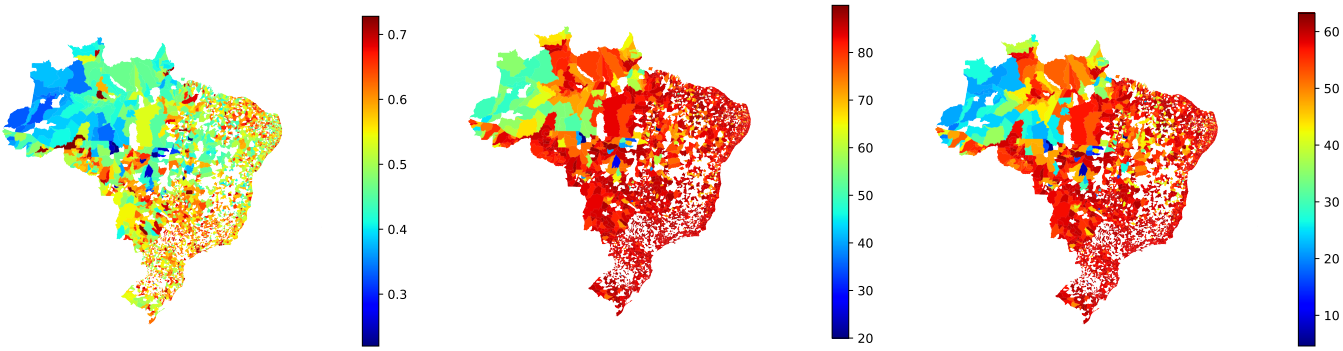


Fig. 2: Histogram of IQS_{leg}, IQS_{rev} and s scores.



(a) Manifold Ranker s (b) $\mathrm{IQS}_{\mathrm{leg}}$ (c) $\mathrm{IQS}_{\mathrm{rev}}$

Fig. 3: Brazilian map visualization of scores for $\mathrm{IQS}_{\mathrm{leg}}$, $\mathrm{IQS}_{\mathrm{rev}}$ and s.